

Supplementary Information for:
*Revisiting Hall–Petch strengthening in FCC
multi-principal-element alloys: Best practices for building
and auditing machine-learning strength models*

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Contents

S1 Dataset audit and exploratory analysis	2
S1.1 Campaign provenance, synthesis, and characterization	2
S1.2 Audited record structure and distributions	3
S1.3 Grouped-validation conventions and primary results	5
S1.4 Why no hard predictive ceiling is reported	6
S1.5 Batch coverage in composition space	6
S2 Family 1: Classical Hall-Petch	7
S3 Family 2: Physics-derived descriptors	11
S3.1 Varvenne–Leyson–Curtin	11
S3.2 Weighted Labusch and Toda–Caraballo adaptations	11
S3.3 Fold-contained PCA	12
S4 Family 3: Composition/processing	13
S5 Family 4: Non-linear ML (ARMOTE-CV)	14
S5.1 Legacy 17-model panel and SHAP audit	15
S6 Family 5: Symbolic regression	16
S6.1 SISSO configuration and expanded controls	16
S6.2 PySR search grid	17
S6.3 Corrected HV fixed form	18
S7 Literature stress test by evidence class	19
S8 Hardness–yield correspondence	20

This Supplementary Information follows the same fixed model-family hierarchy as Fig. 1 of the main paper: Family 1 (Classical Hall-Petch), Family 2 (Physics-derived descriptors), Family 3 (Composition/processing), Family 4 (Non-linear ML (ARMOTE-CV)), and Family 5 (Symbolic regression). The names and order remain unchanged in both documents. S1–S4 are cumulative input sets within Family 4, not separate model families. E1–E3 are literature evidence classes and are also separate from the model hierarchy.

Primary internal performance is reported using shuffled 5-fold cross-validation, leave-one-out (LOO), and leave-one-batch-out (LOBO) validation. LOO operates at the record level. When a composition is repeated, another record with the same composition can remain in the training set. We therefore do not describe LOO as strict unseen-composition validation. LOBO tests transfer to a new Bayesian-optimization iteration. External literature testing and singularity audits of closed-form equations are reported separately from internal predictive validation.

S1 Dataset audit and exploratory analysis

S1.1 Campaign provenance, synthesis, and characterization

The broader alloy-discovery effort began with Campaign A, which established the closed-loop, multi-objective workflow [1]. Campaign A is a methodological prequel and does not contribute records to the present property-modeling dataset. This dataset combines the subsequent B and C Bayesian-optimization campaigns carried out at Texas A&M University through the BIRDSHOT initiative within the ARL HTMDEC program [2, 3]. The B campaign contributes 35 records from batches BBA, BBB, and BBC, including 34 with YS. It explored the Al–Co–Cr–Cu–Fe–Mn–Ni–V system under an FCC phase-stability constraint while holding processing fixed. Section 2.2 of Ref. [3] documents the B-campaign synthesis and specimen-preparation workflow.

The C campaign contributes 59 records from CBA, CBB, and CBC. It used insights from the B campaign to focus on V–Cr–Mn–Fe–Co–Ni alloys and jointly varied composition, cold work, recrystallization temperature, and hold time. During design-space curation, a modified JMAK-type static-recrystallization and grain-growth prior provided grain-size estimates for feasible candidates. The complete C-campaign and grain-growth-model details will be reported separately. The present regression analysis uses measured EBSD grain sizes rather than the design-stage estimates.

Both campaigns followed the same high-level route of vacuum-arc melting from high-purity elements, homogenization, cold rolling, recrystallization, and quenching. EDS and XRD checked realized chemistry and phase constitution; SEM and EBSD documented the microstructure; and quasistatic tensile and Vickers indentation measurements supplied YS and HV. Figure 2 of the main paper shows representative chemistry and microstructure verification. Figure S1 shows the non-uniform composition and processing coverage that motivates grouped validation.

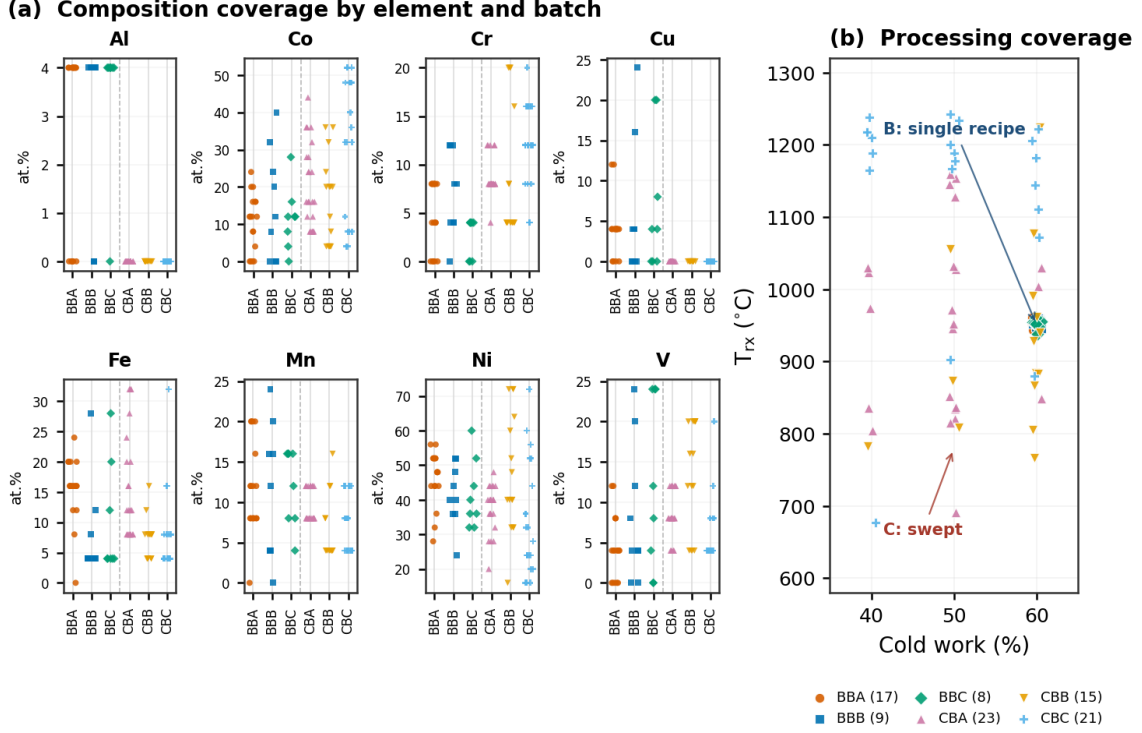


Figure S1: Composition and processing coverage by batch. The B campaign uses one fixed processing condition, while the C campaign spans a processing grid. The coverage follows the experimental design and is not a random sample of the composition simplex.

S1.2 Audited record structure and distributions

The source workbook contains 94 records and 19 original columns. Every composition row sums to 100 at.% (Table S1). YS is available for 93 records, and HV is available for all 94. BBA09 is the only record without YS. The B batches contain 35 hardness records and 34 YS records. The C batches contain 59 records for both targets. Processing is fixed within the B campaign, whereas the C campaign varies cold work, recrystallization temperature, and hold time together with composition.

Table S1: Audited dataset statistics. Means and standard deviations are sample statistics.

Quantity	n	Min	Max	Mean $\pm$ SD
YS (MPa)	93	151.5	544.5	268.5 $\pm$ 81.6
HV	94	57.6	227.5	133.9 $\pm$ 33.6
d (μm)	94	14.66	211.75	57.39 $\pm$ 42.67
SD_{grain} (μm)	94	8.32	321.06	44.72 $\pm$ 44.83

The full dataset contains 82 unique compositions. The 93-record YS subset contains 81. Nine composition groups are repeated and account for 21 records. One composition, $\text{Co}_4\text{Cr}_4\text{Fe}_4\text{Mn}_4\text{Ni}_{72}\text{V}_{12}$, appears in both CBB and CBC, with three CBB records and one CBC record. Record-level LOO can therefore contain exact-composition overlap. LOBO reduces this overlap but does not remove it completely. We report these facts as part of the validation audit rather than treating them as another validation protocol.

Mean grain size and its standard deviation are strongly correlated ($r = 0.801$). The dimensionless coefficient of variation, SD_{grain}/d , ranges from 0.47 to 2.99. A claim that distribution

width has an independent physical role should therefore compare raw SD with normalized width and with distribution moments such as $E[d^{-1/2}]$.

The source data also contain measurement scatter, which is distinct from grain-distribution width. SD_{YS} is available for 80 records and ranges from 0.2 to 98.7 MPa; SD_{HV} is available for all 94 and ranges from 1.43 to 12.44 HV. These values are not used as model features because replicate counts and reporting definitions are not consistent across records. In contrast, SD_{grain} is an EBSD microstructure descriptor and is evaluated explicitly as a predictor.

The strongest composition–grain-size correlations include $r(d, c_{\text{Cr}}) = +0.38$ and $r(d, c_{\text{Al}}) = -0.32$. On average, Cr-rich alloys are therefore coarser and Al-rich alloys are finer. These signs were reversed in the earlier draft. A regression of $d^{-1/2}$ on composition explains approximately 40% of its variance. Adding processing raises this value to approximately 54%. Composition and grain size are therefore not experimentally orthogonal.

Figure S2 shows the global property distributions. The gap in the grain-size histogram comes from the C-campaign processing grid and the B campaign’s fixed recipe. It is not evidence of a bimodal grain-growth mechanism. Figure S3 shows the C-campaign true stress–strain curves used to extract 0.2% offset YS. Figure S4 summarizes correlations among composition, processing, and descriptors. Several deterministic descriptors are strongly correlated with one another, so coefficient and feature-importance interpretations must account for redundancy.

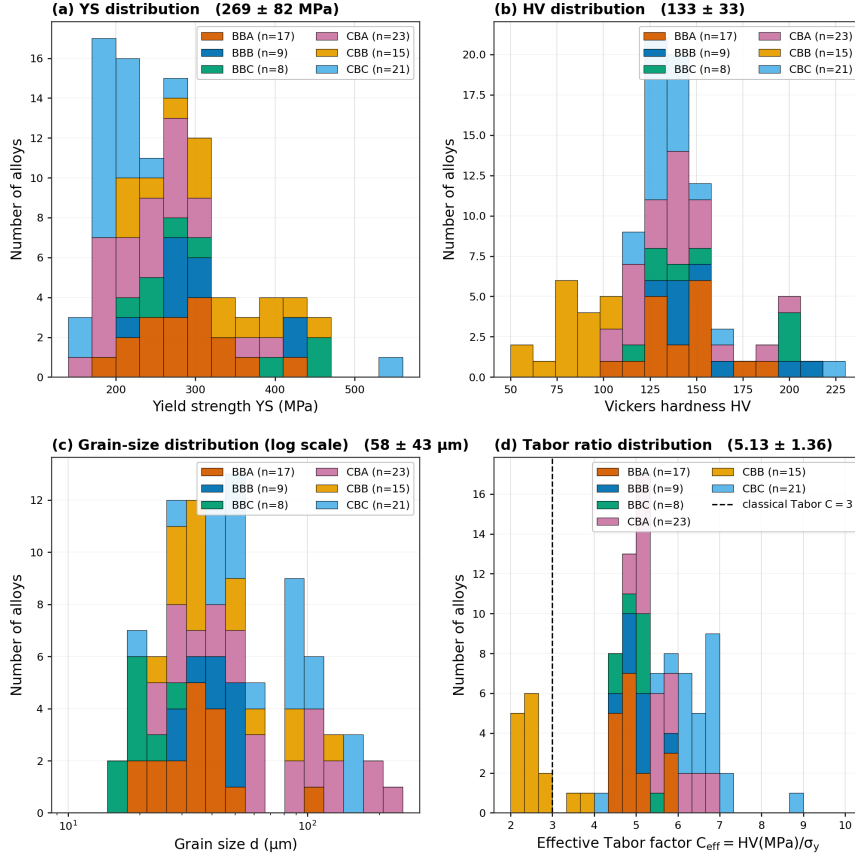


Figure S2: Audited YS, HV, grain-size, and effective hardness-to-yield distributions by batch.

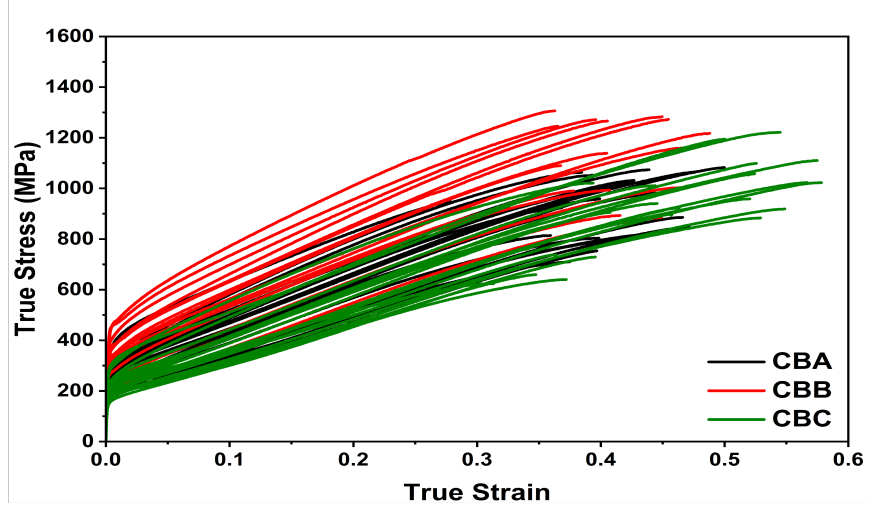


Figure S3: True stress-strain curves for the C-campaign records at a nominal strain rate of $5 \times 10^{-4} \text{ s}^{-1}$. The 0.2% offset values are the YS targets used in the analysis.

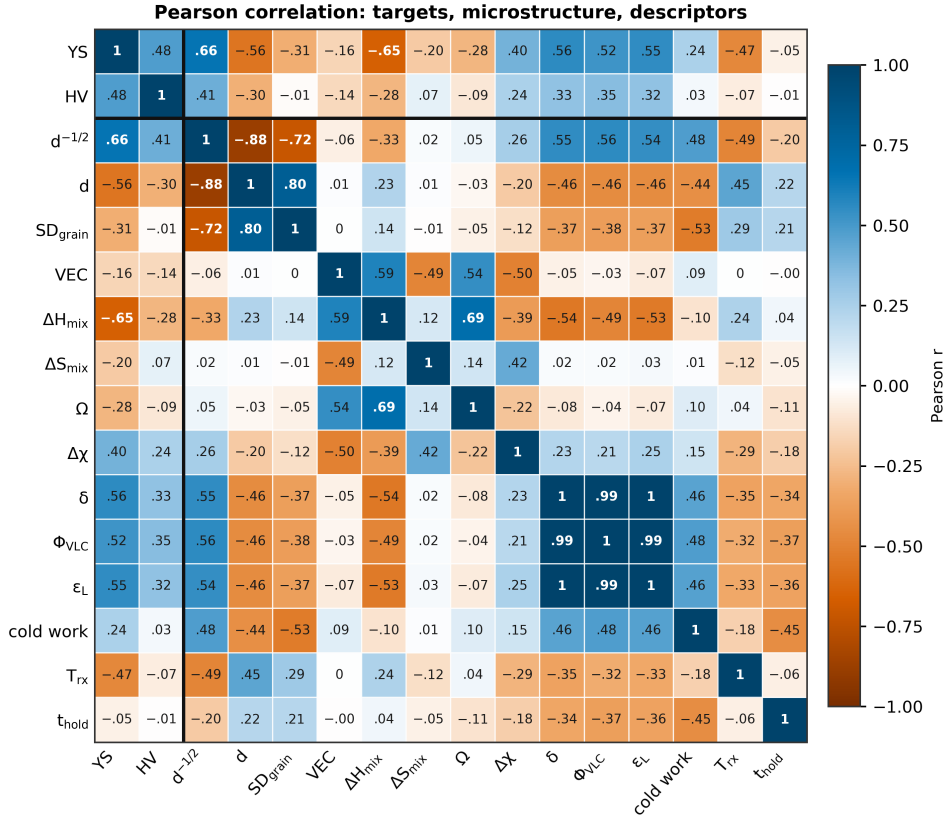


Figure S4: Pearson correlation matrix for composition, processing, grain-size, and descriptor variables. Correlation is descriptive and does not establish independent mechanistic effects.

S1.3 Grouped-validation conventions and primary results

For every protocol, predictions from all held-out records are pooled before calculating

$$Q^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (\text{S1})$$

The manuscript reports this pooled quantity as cross-validated R^2 . It is not the mean of the fold-specific R^2 values; Table S2 lists the pooled scores for every model. Preprocessing is fit within each training fold. To calculate batch-clustered intervals, the six batches are resampled with replacement 1000 times while each batch remains intact. Because there are only six clusters, these intervals show sensitivity to the sampled batches but are not precise inferential confidence intervals.

Table S2: Primary grouped-validation results reproduced by the revision-specific validation script. The Family 5 HV result is conditional on an equation structure selected using the complete dataset before the coefficients were refit in each fold.

Target	Family/model	Status	5-fold	LOO	LOBO
YS	F1 Classical HP	baseline	0.405	0.406	0.373
YS	F2 PCA-OLS	fold-contained	0.462	0.480	0.362
YS	F3 M3	pre-specified	0.666	0.652	0.625
YS	F3 M15 SD interaction	pre-specified	0.731	0.694	0.694
YS	F4 Lasso S2	fixed settings	0.683	0.674	0.621
YS	F4 LightGBM S2	fixed settings	0.632	0.574	0.615
HV	F1 Classical HP	baseline	0.086	0.136	-0.077
HV	F4 LightGBM S1	fixed settings	0.195	0.323	0.112
HV	F5 fixed form	post-selected	0.725	0.727	0.636

The validation hierarchy changes the scientific interpretation. LOO measures record-level interpolation and can benefit from repeated or nearby compositions. LOBO tests a new iteration while retaining the practices shared across the broader program, and is therefore the primary internal grouped-transfer test. The literature stress test asks a different question about numerical behavior on heterogeneous published records, while singularity auditing checks whether a closed form remains finite and bounded over the audited domain.

S1.4 Why no hard predictive ceiling is reported

The source analysis contained two incompatible ceiling calculations. If every singleton composition is assigned zero residual, an in-sample composition-mean oracle gives approximately $R^2 = 0.86$. If variance is pooled only from repeated compositions, the within-composition variance is approximately 6979 MPa². This is slightly larger than the total YS variance of 6651 MPa², giving a formal value near -0.05 . Neither value is a defensible out-of-sample ceiling. There are 72 singleton YS compositions, and the repeated groups also differ in grain size and processing.

Similarly, $1 - \langle \text{SD}_{YS}^2 \rangle / \text{Var}(YS) \approx 0.97$ is not a universal noise ceiling. This calculation assumes that the reported within-alloy standard deviations are irreducible errors in the modeled target. If the target is a replicate mean, its sampling uncertainty depends on the number of replicates through $\text{SD}/\sqrt{n}$. A measurement-limited ceiling cannot be estimated without replicate-level measurements and replicate counts.

S1.5 Batch coverage in composition space

Six principal components are needed to capture approximately 95% of the composition variance. In the PC1–PC2 projection, the mean pairwise convex-hull overlap is approximately 0.21 (Fig. S5). This two-dimensional diagnostic does not prove support in the full composition space, but it explains why the LOBO folds differ in difficulty. CBC is relatively isolated, while CBA is well covered by the remaining batches. The aggregate LOBO score therefore combines near-interpolation with genuine extrapolation. It should be read together with the per-batch results in Table S13.

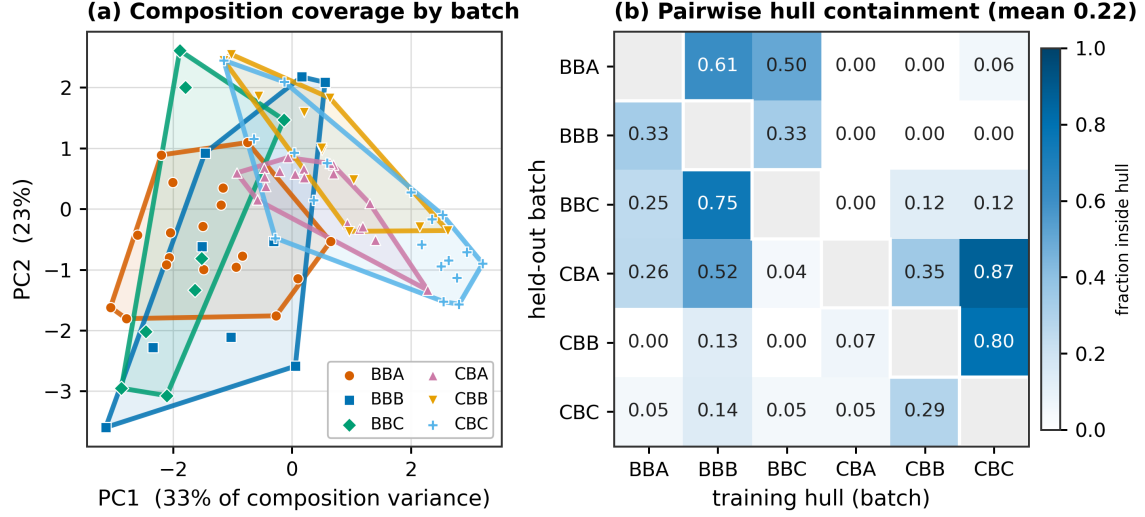


Figure S5: Batch coverage in the PC1–PC2 projection of composition space: batch convex hulls (left) and pairwise hull containment (right). These values depend on the projection and do not guarantee support in the full composition space.

S2 Family 1: Classical Hall-Petch

Nine pre-specified scaling laws were fit to YS (Table S3). Five monotonic laws fall within $\Delta\text{BIC} < 2$ of the classical Hall–Petch form. The optimized exponent is $n = 0.548$. Its Bayesian posterior is 0.53 ± 0.16 , with a 94% HDI of approximately $[0.22, 0.82]$. These results support $d^{-1/2}$ as a simple baseline but do not statistically rule out nearby exponents.

Table S3: Family 1 comparison of pre-specified grain-size laws for YS, ranked by BIC. Here, p is the number of fitted regression coefficients.

Scaling law $f(d)$	p	Train R^2	LOO R^2	LOO RMSE	BIC	ΔBIC
$d^{-1/2}$ (Hall–Petch)	2	0.429	0.406	62.5	774.5	0.0
$\ln(d)/d$	2	0.429	0.406	62.5	774.5	0.0
$d^{-2/3}$	2	0.429	0.405	62.6	774.6	0.1
$d^{-1/3}$	2	0.427	0.403	62.6	774.9	0.4
d^{-1}	2	0.419	0.395	63.1	776.1	1.6
$\ln(d)$	2	0.413	0.388	63.4	777.2	2.7
$d^{-0.548}$	3	0.430	0.406	62.5	779.0	4.5
$d^{-1/2} + d^{-1}$	3	0.430	0.395	63.1	779.0	4.5
$d^{-1} + d^{-2}$	3	0.429	0.387	63.5	779.1	4.6

Table S4: Bayesian PSIS-LOO comparison for the Family 1 YS laws. The stacking solution is reported but is not treated as proof of a unique law.

Model	elpd_{loo}	ΔLOO	p_{loo}	Weight
Hall-Petch $d^{-1/2}$	-518.5	0.0	4.2	1.000
Composite $d^{-1/2} + d^{-1}$	-518.5	0.1	4.2	0.000
Critical-thickness $\ln(d)/d$	-518.5	0.1	4.3	0.000
Intermediate $d^{-2/3}$	-518.7	0.2	4.4	0.000
Baldwin $d^{-1/3}$	-518.7	0.2	4.2	0.000
Optimized exponent d^{-n}	-518.8	0.3	4.5	0.000
Logarithmic $\ln(d)$	-519.7	1.2	3.9	0.000
Dunstan-Bushby d^{-1}	-520.3	1.8	4.3	0.000

The classical YS baseline gives LOO/LOBO $R^2 = 0.406/0.373$ (Fig. S6). For HV, the corresponding values are $0.136/-0.077$. Mean grain size is therefore a useful YS baseline within the sampled program, but it does not transfer reliably across batches for HV.

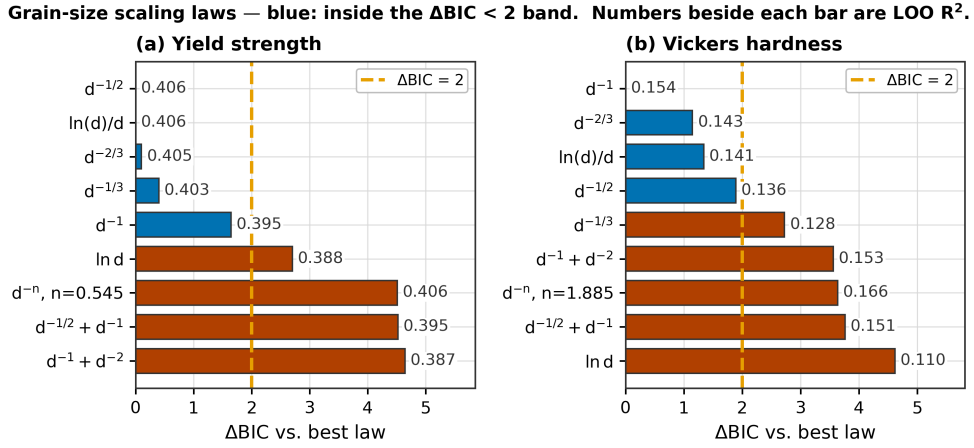


Figure S6: Information-criterion differences among pre-specified YS grain-size laws.

The Bayesian comparison is summarized in Table S4, with the fitted laws in Fig. S7, the free-exponent posterior in Fig. S8, and the model weights in Fig. S9. The Bayesian stacking weight concentrates on the classical form, even though several candidate laws have almost identical expected log predictive density. We therefore do not treat this boundary solution as decisive evidence for one exponent. Parameter counting also differs across the stored tables. Some Bayesian tables include the noise parameter, while the frequentist BIC tables count regression coefficients. The revised manuscript compares models only within internally consistent tables.

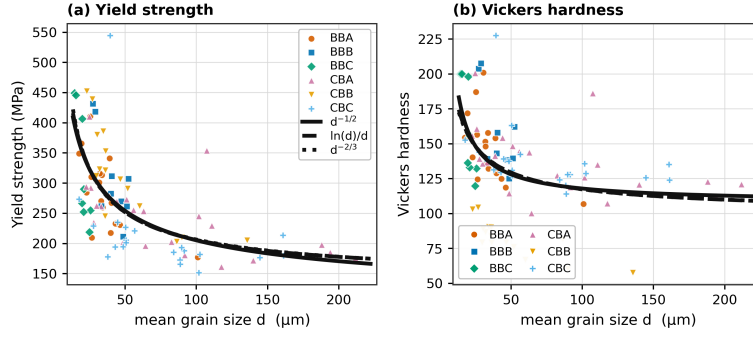


Figure S7: Top Family 1 YS scaling laws over the observed grain-size range. Their close overlap shows why the data do not identify a unique exponent.

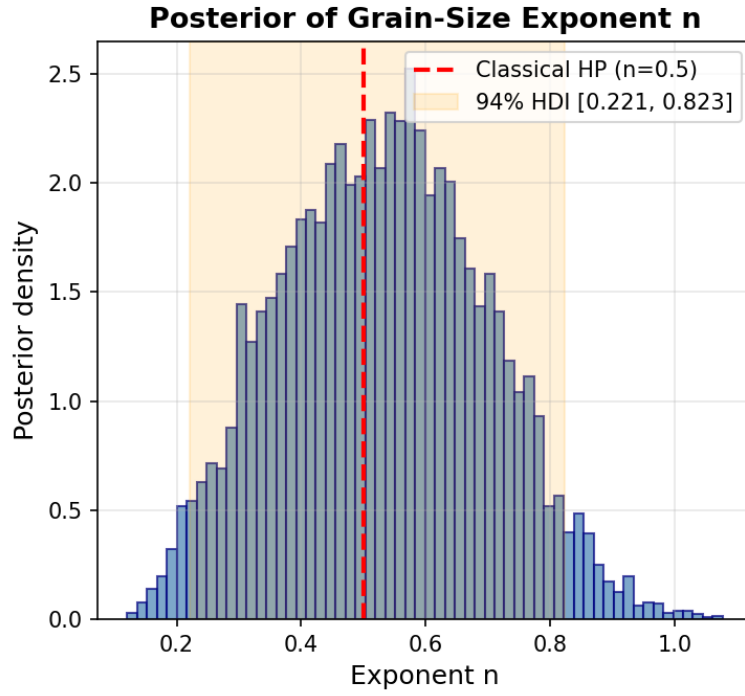


Figure S8: Posterior distribution of the free grain-size exponent n in $\sigma_y = \sigma_0 + kd^{-n}$; the dashed line marks $n = 0.5$.

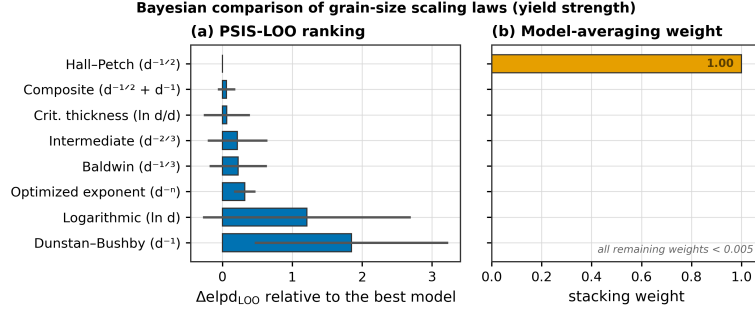


Figure S9: Bayesian model-averaged grain-only prediction with its 94% credible band. The large residual scatter shows that grain size alone does not capture the dominant property variation.

The within-replicate analysis (Table S5) gives local slopes from approximately -8539 to 9282 in Hall-Petch units. The fixed-effect pooled slope is near 2144 , with Cochran $Q \approx 251$ and $I^2 = 96.8\%$. This apparent heterogeneity cannot be attributed to composition because most groups contain only two records and processing is not held fixed. In addition, the two-stage k_{eff} regression subtracts an intercept from a model that already assumes constant k . The supported conclusion is that constant-slope M3 is the best predictive model in Family 3. This design cannot identify variation in k_{HP} .

Table S5: Descriptive within-composition fits for the nine repeated-composition groups (21 records). The values compare transformations after group demeaning but do not identify composition-specific slopes.

Scaling law	Pooled within-group R^2	R^2 for the $n = 3$ group
$d^{-1/2}$	0.778	0.995
$d^{-1/3}$	0.781	0.997
$d^{-2/3}$	0.775	0.993
d^{-1}	0.769	0.987
$\ln(d)/d$	0.776	0.993
$\ln d$	0.786	0.999

Table S6: Family 1 grain-size laws fit separately to HV. The optimized exponent was selected using the full dataset and is descriptive.

Scaling law $f(d)$	p	Train R^2	LOO R^2	ΔBIC
$d^{-1.728}$	3	0.193	0.166	0.0
d^{-1}	2	0.185	0.154	0.9
$d^{-2/3}$	2	0.175	0.143	2.1
$\ln(d)/d$	2	0.174	0.141	2.3
$d^{-1/2}$	2	0.169	0.136	2.8
$d^{-1/3}$	2	0.161	0.128	3.7
$d^{-1/2} + d^{-1}$	3	0.192	0.151	4.7
$\ln(d)$	2	0.144	0.110	5.6

S3 Family 2: Physics-derived descriptors

S3.1 Varvenne–Leyson–Curtin

Following Varvenne et al. [4, 5], the zero-temperature critical resolved shear stress is

$$\tau_{y0} = 0.051\alpha^{-1/3}\bar{\mu}\left(\frac{1+\bar{\nu}}{1-\bar{\nu}}\right)^{4/3}f_1(w_c)\left(\frac{\sum_i c_i\Delta V_i^2}{b^6}\right)^{2/3}, \quad (\text{S2})$$

and the finite-temperature polycrystal prediction is

$$\sigma_{\text{VLC}}(T) = M\tau_{y0}\left[1 - \left(\frac{k_B T}{\Delta E_b}\right)^{2/3}\right]. \quad (\text{S3})$$

The activation barrier is

$$\Delta E_b = 0.274\alpha^{1/3}\bar{\mu}b^3\left(\frac{1+\bar{\nu}}{1-\bar{\nu}}\right)^{2/3}f_2(w_c)\left(\frac{\sum_i c_i\Delta V_i^2}{b^6}\right)^{1/3}. \quad (\text{S4})$$

Here $M = 3.06$, $\alpha = 0.123$, $f_1 = 0.35$, $f_2 = 5.70$, and $b = (4\bar{V})^{1/3}/\sqrt{2}$. We use binary-solid-solution-derived volumes for Ni–Co–Fe–Cr–Mn and tabulated or extrapolated FCC volumes for Al, Cu, and V. Rule-of-mixtures elastic constants replace measured alloy values. These choices are material approximations and do not exactly reproduce a first-principles VLC calculation.

S3.2 Weighted Labusch and Toda–Caraballo adaptations

The weighted Labusch extension is

$$\sigma_{\text{Lab}} = \bar{\mu}\varepsilon_L^{4/3}c_{\text{eff}}^{2/3}, \quad \varepsilon_L = \sqrt{\sum_i c_i(\delta_{\mu,i}^2 + \alpha_L^2\delta_{r,i}^2)}, \quad (\text{S5})$$

with $\alpha_L = 16$. This multicomponent weighting is an adaptation of Labusch theory [6]. The Toda–Caraballo implementation is

$$\sigma_{\text{TC}} = M\left(\sum_i B_i^{3/2}X_i\right)^{2/3}, \quad B_i = 3\bar{\mu}\varepsilon_i^{4/3}Z, \quad (\text{S6})$$

with damped modulus and Vegard size misfits [7]. Pair-specific binary B_i values are not available for the eight-element system, so we use a shared theoretical prefactor.

Table S6 repeats the scaling-law comparison for hardness. The Labusch and Toda–Caraballo constants are calibrated once to reproduce 125 MPa for CoCrFeMnNi [8]. VLC is not calibrated. Table S8 separates the calibrated errors from the $2\text{--}5\times$ errors of the uncalibrated variants.

Table S7: Cantor-anchored SSS benchmark for source alloys with traceable references in the local bibliography. Values are model outputs in MPa. $\sigma_{0,\text{exp}}$ is the reported YS after the cited Hall–Petch contribution is subtracted.

Alloy	VLC	Labusch	Toda–Caraballo	$\sigma_{0,\text{exp}}$
NiCoCr [9]	233	233	135	218
NiCoFeCr [10]	222	160	146	100
CoCrFeMnNi [8]	243	125	125	125

Table S8: Family 2 SSS predictions compared with Family 1 Hall–Petch-corrected YS for 93 alloys. Ratio is the mean prediction divided by the mean corrected response. +HP uses Ridge with the SSS estimate and $d^{-1/2}$.

Model	Mean	Ratio	Raw R^2	Spearman ρ	+HP LOO R^2	RMSE
VLC	342	1.78	−6.92	0.53	0.396	63.1
Labusch, calibrated	206	1.07	0.14	0.55	0.469	59.1
Toda–Caraballo, calibrated	274	1.42	−3.18	0.49	0.399	62.9
HP only	—	—	—	—	0.406	62.5

Table S7 lists the benchmark, and Fig. S10 shows the parity plots. As standalone LOO predictors, the archived SSS scores are 0.274 for calibrated Labusch, 0.091 for Toda–Caraballo, and 0.030 for VLC. All three are below the Family 1 YS baseline. When raw composition is already included, adding one of these deterministic SSS estimates changes LOO R^2 by no more than 0.002. The reported linear partial correlations after removing the composition signal are also below $|r| = 0.10$. These results show that the estimates add little information in this dataset. They do not invalidate the underlying theories outside the present input approximations.

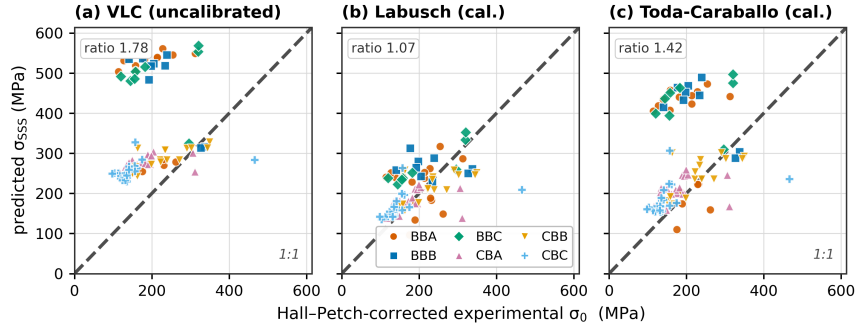


Figure S10: Family 2 SSS estimates versus Family 1 Hall–Petch-corrected YS.

The present data cannot show that the error comes only from the Vegard inputs. Approximated elastic properties, calibration transfer, phase or local-order effects, and the subtraction used to define experimental friction stress can all contribute. DFT-derived misfit volumes improve VLC predictions in selected FCC systems [11]. Improved inputs are therefore a clear next test, not a proven single explanation.

S3.3 Fold-contained PCA

The earlier Family 2 PCA result fit StandardScaler and PCA on all 93 records before CV (Table S9). In the revised calculation, both transformations are fit within every training fold. The model uses 11 inputs: six Wen descriptors, $d^{-1/2}$, SD_{grain} , cold work, recrystallization temperature, and hold time.

Table S9: Family 2 PCA-OLS before and after correcting preprocessing leakage.

Implementation	5-fold R^2	LOO R^2	LOBO R^2
Global scaling/PCA	0.525	0.516	0.441
Fold-contained scaling/PCA	0.462	0.480	0.362

PCA loadings from the full dataset remain descriptive only. The back-projected coefficients depend on feature scaling and are not interpreted as causal feature importance. The global-

preprocessing implementation is retained only to document leakage and is not used for the revised comparison.

S4 Family 3: Composition/processing

Family 3 contains the OLS and M-model Hall–Petch hierarchy. These models add raw composition, recorded processing, and selected microstructure terms to the Family 1 baseline. M3 uses a shared k_{HP} while allowing the intercept to depend on composition. M15 extends M3 with additive SD_{grain} and the interaction $\text{SD}_{\text{grain}}d^{-1/2}$. Their coefficients are empirical within the sampled campaigns and are not elemental SSS constants.

M3 gives LOO/LOBO $R^2 = 0.652/0.625$. The modest change from LOO to LOBO supports transfer among the six sampled iterations. Because LOO can retain a repeated composition, we do not use it to claim strict prediction of unseen compositions.

Table S10 separates the value of measuring grain-distribution width from the value of its interaction with the Hall–Petch term. Adding SD_{grain} only as an independent term improves LOO but lowers LOBO. M15 improves both and is the strongest verified low-complexity YS model when distribution width is available. The interaction coefficient is 19.16 in the fitted numerical units, with $t = 4.71$ and $p = 1.0 \times 10^{-5}$. Relative to M3, M15 lowers BIC by 20.1. These are reproducible OLS results, but they do not identify a causal grain-width mechanism because d , SD_{grain} , composition, and processing are correlated.

Table S10: Verified Family 3 grain-width controls. Values are pooled out-of-fold R^2 .

Model	Parameters	5-fold	LOO	LOBO
M3	9	0.666	0.652	0.625
M3 + additive SD_{grain}	10	0.659	0.668	0.595
M15 + SD–HP interaction	11	0.731	0.694	0.694

Table S11: Family 3 M3 coefficients relative to Ni.

Parameter	Estimate (MPa)	95% low	95% high
Intercept	229.6	152.4	306.7
Al	−201.3	−955.4	552.8
Co	−187.2	−268.1	−106.3
Cr	−307.7	−588.5	−26.9
Cu	−334.0	−651.9	−16.2
Fe	−360.4	−520.1	−200.6
Mn	82.3	−117.6	282.2
V	291.3	41.0	541.5
k_{HP}	765.8	501.5	1030.0

The fitted M3 coefficients are listed in Table S11, and literature Hall–Petch slopes for comparison in Table S12. Al has a much wider interval than the named stable coefficients because it spans only 0–4 at.%. We therefore avoid stating that every M3 coefficient changes by less than $\pm 15\%$ under grain-size perturbation. That diagnostic applies specifically to V, Co, Cr, and Fe. In 1000 grain-size perturbations, the V, Co, Cr, and Fe coefficients changed by less than approximately 15%, while k_{HP} changed by approximately 39%. A 10,000-replicate row bootstrap gave standard-error ratios of 0.99–1.54 relative to OLS. These checks support numerical stability within the sampled records. However, the row bootstrap does not reproduce campaign-level sampling and does not guarantee transfer.

Table S12: Literature context for the fitted constant Hall–Petch slope. The table includes only the directly cited FCC entries retained from the source supplement.

Alloy	k_{HP} (MPa $\cdot\mu\text{m}^{1/2}$)	Source
Pure Cu	110	Cordero et al. [12]
Pure Ni	160	Cordero et al. [12]
CoCrNi	265	Yoshida et al. [9]
CoCrFeMnNi	494	Otto et al. [8]
CoCrFeNi	516	Yoshida et al. [13]
This work, Family 3 M3	766	shared fitted slope

The grain-only Family 1 slope is approximately 1149 MPa $\cdot\mu\text{m}^{1/2}$. Adding the composition-dependent intercept in M3 reduces the shared slope to approximately 766 MPa $\cdot\mu\text{m}^{1/2}$. This change is consistent with composition–grain-size confounding in the global baseline. It does not show that 766 is a universal slope for FCC MPEAs.

Table S13: Family 3 M3 per-batch LOBO performance and composition-hull containment. The aggregate LOBO value in the main text is pooled; it is not the mean of these R^2 values.

Batch	n	Hull	F1 HP R^2	F3 M3 R^2	M3 RMSE
BBA	17	0.882	0.412	0.697	32.9
BBB	9	0.556	0.138	0.306	57.7
BBC	8	0.750	0.259	0.656	51.9
CBA	23	1.000	0.295	0.423	44.9
CBB	15	0.933	0.174	0.584	49.6
CBC	21	0.476	−0.038	0.409	60.4

S5 Family 4: Non-linear ML (ARMOTE-CV)

The equal-footing panel gives every estimator the same S1–S4 input matrices and fixed splits. S1 contains both $d^{-1/2}$ and SD_{grain} , unlike the Family 1 classical baseline, which contains only $d^{-1/2}$. For YS, an S1 linear model gives LOO/LOBO $R^2 = 0.451/0.394$, compared with 0.406/0.373 for Family 1. The larger M15 gain therefore comes from the interaction, not simply from adding the SD feature. At S2, the Lasso linear control and LightGBM have similar LOBO values of 0.621 and 0.615. For HV, adding SD linearly changes LOO/LOBO from 0.136/−0.077 to 0.232/−0.176, while LightGBM S1 gives 0.323/0.112. Thus the LightGBM-to-classical comparison changes both features and estimator. Descriptor-rich input sets are generally negative under LOBO. These are comparisons within a shared Family 4 panel; they do not imply that every repository analysis uses identical inputs.

The primary audit in the main paper uses declared, fixed settings with no data-driven hyperparameter selection. Lasso uses $\alpha = 1.0$. LightGBM uses 400 estimators, a maximum depth of 3, a learning rate of 0.05, and a feature fraction of 0.8. These settings were chosen for this study and are not library defaults. The batch-bootstrap LOBO intervals overlap for YS Lasso ([0.516, 0.664]) and LightGBM ([0.474, 0.664]). This overlap does not establish equivalence, but it provides no evidence that non-linearity improves the matched S2 comparison.

The archived nested ARMOTE-CV summary reports YS Bayesian ridge at S2 with $R^2 = 0.685$ under 5-fold CV, 0.666 under LOO, and 0.613 under LOBO. Its generator and raw fold objects are not in the repository. The available `armote_ladder_cv.py` script tunes once on the full dataset and then evaluates the selected settings. It therefore does not verify the archived nested implementation. We retain the archived Family 4 summary as secondary evidence until the fold records are released.

S5.1 Legacy 17-model panel and SHAP audit

The legacy 17-model panel tuned selected models on the full dataset before LOO. It also gave the tree models up to 64 interaction features. Its XGBoost scores, 0.729 under LOO and 0.574 under LOBO, are therefore not confirmatory. The 0.155 difference is a validation gap; it does not prove that batch patterns cause 21% of the explained variance. Table S14 is retained for provenance. It should not be compared numerically with the fixed-setting primary panel as though both used the same selection protocol.

Table S14: Archived legacy panel ranked by LOO R^2 . INT-ALT is a 64-feature interaction matrix, and COMPACT is a 20-feature matrix selected using the full dataset. Effective parameter counts and BIC are omitted because their definitions differed across estimator classes.

Model	Features	p	LOO R^2	LOBO R^2
XGBoost	INT-ALT	64	0.729	0.574
Average ensemble	meta	5	0.700	—
Stacking (Ridge)	meta	5	0.698	0.707
Random forest	INT-ALT	64	0.675	0.623
LightGBM	INT-ALT	64	0.673	0.625
SVR (RBF)	COMPACT	20	0.654	0.497
Family 3 M3	HP-PHYS	9	0.652	0.625
XGBoost compact	COMPACT	20	0.650	0.580
Ridge	PHYS-ALT	33	0.638	0.651
ElasticNet	PHYS-ALT	33	0.637	0.648
CatBoost	INT-ALT	64	0.637	0.515

The stacking result has an additional leakage risk. Its meta-features are LOO predictions from base models that retained other records from the held-out batch. In the legacy XGBoost model, SHAP ranks Ω , $\text{Mn} \times d^{-1}$, $\text{V} \times d^{-1}$, alternative grain-size transforms, and SSS descriptors highly (Fig. S11). These attributions describe the archived fitted model. Because the features are correlated and include constructed interactions, the SHAP values cannot identify independent physical mechanisms or a composition-dependent Hall–Petch slope.

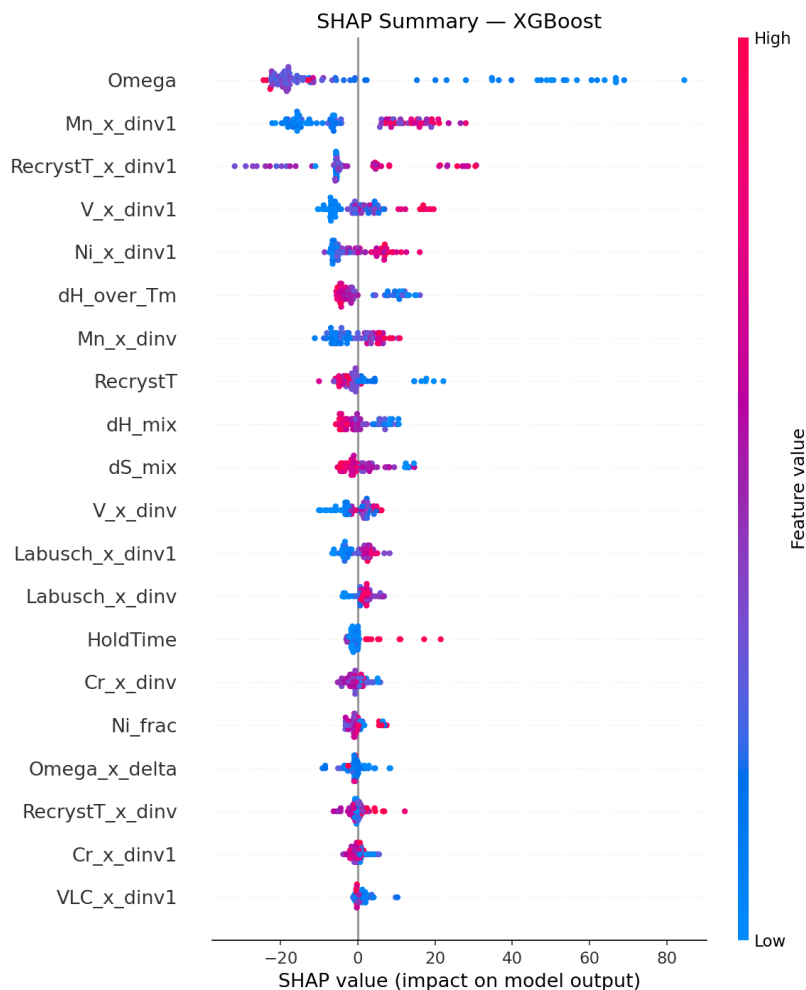


Figure S11: SHAP summary for the archived XGBoost model. The values explain this predictor, which was tuned using the full dataset; they are not causal feature effects.

S6 Family 5: Symbolic regression

S6.1 SISSO configuration and expanded controls

The archived SISSO input space contains 27 primary features. These include composition-weighted statistics of atomic radius, shear modulus, electronegativity, and melting point; thermodynamic descriptors; three SSS-related quantities; and $d^{-1/2}$ [14, 15]. Two algebraic expansion tiers using $+$, $-$, $\times$, and $\div$ are followed by sparse selection at descriptor dimension $D = 3$. The “SISSO Full” expression was selected from this space. The “SISSO Robust” control removes the small-denominator δ_μ term before the search. Here, “Robust” means numerically bounded over the audited domain, not validated transfer.

An expanded archived search added x^2 , $\sqrt{x}$, and x^{-1} , nine alternative grain-size transforms, a larger screening threshold, and BIC selection over $D = 1, \dots, 4$. Its reported LOO/LOBO $R^2 = 0.604/0.350$ was lower than the constrained-search result because the selected transforms changed across folds. An EML universal-operator control [16] gave LOO $R^2 = 0.321$ at depth one and became unstable at greater depth. A recalibrated pure-metal descriptor relation [17] gave LOO $R^2 = 0.346$. These controls support the hypothesis that a useful inductive bias can help. However, the archived records do not show that every structure-selection decision was nested within each outer split.

S6.2 PySR search grid

PySR used three feature pools, renamed P1–P3 here to avoid confusion with the five Family labels. P1 contains $d^{-1/2}$ and SD_{grain} . P2 adds raw composition and processing. P3 instead adds six Wen descriptors and processing. The archived files call these pools F1–F2. The operator sets were O1 (+, −), O2 (+, −, ×, ÷), and O3 (O2 plus protected square root, logarithm, square, and cube). Each cell used 50–75 search iterations. Equation structures and elbow or accuracy selections were obtained using the complete target dataset.

The full PySR grid is given in Table S15 and its complexity–loss front in Fig. S12. For CV, the numerical constants in the selected full-data expression were replaced with free parameters and refit in each training fold. This tests a frozen structure after it has been selected; it does not repeat the structure search. The stored LOBO values in Table S16 are averages of per-batch R^2 . In contrast, the confirmatory values in the main text are pooled Q^2 . We report the stored values for provenance but do not place them on the same leaderboard.

Table S15: Archived PySR grid selected at the elbow. Stored LOBO uses the archived fold aggregation. Ext. is RMSE on the heterogeneous E1–E3 literature stress set, and Safe records the archived singularity audit. P1–P3 are feature pools within Family 5.

Target	Cell	Const.	LOO R^2	Stored LOBO	BIC	Ext. RMSE	Safe
YS	P1–O1	1	0.019	−0.251	818	112	yes
YS	P1–O2	2	0.482	0.276	761	168	yes
YS	P1–O3	1	0.399	0.205	773	158	yes
YS	P2–O2	2	0.605	0.475	737	114	yes
YS	P2–O3	2	0.604	0.496	737	108	yes
YS	P3–O1	1	0.290	0.068	788	117	yes
YS	P3–O2	2	0.632	0.500	727	133	yes
YS	P3–O3	2	0.674	0.555	716	128	yes
HV	P1–O1	1	−0.018	−4.16	664	32	yes
HV	P1–O3	3	0.579	−1.15	587	9944	no
HV	P2–O2	3	0.568	−1.23	587	97	yes
HV	P2–O3	4	0.678	−0.48	562	17895	no
HV	P3–O2	3	0.631	−0.87	570	75	yes
HV	P3–O3	4	0.681	−2.79	561	51	yes

Table S16: Selected PySR YS equations. CV values are post-selection fixed-form scores. Stored LOBO is the mean of the fold-wise R^2 values. Every external R^2 is negative.

Cell	Selection	LOO R^2	Stored LOBO	Ext. RMSE	Ext. R^2
P2–O2	accuracy	0.808	0.696	129.4	−0.496
P2–O3	accuracy	0.808	0.689	111.9	−0.119
P3–O3	accuracy	0.765	0.668	120.4	−0.295
P3–O3	elbow	0.674	0.555	128.4	−0.474

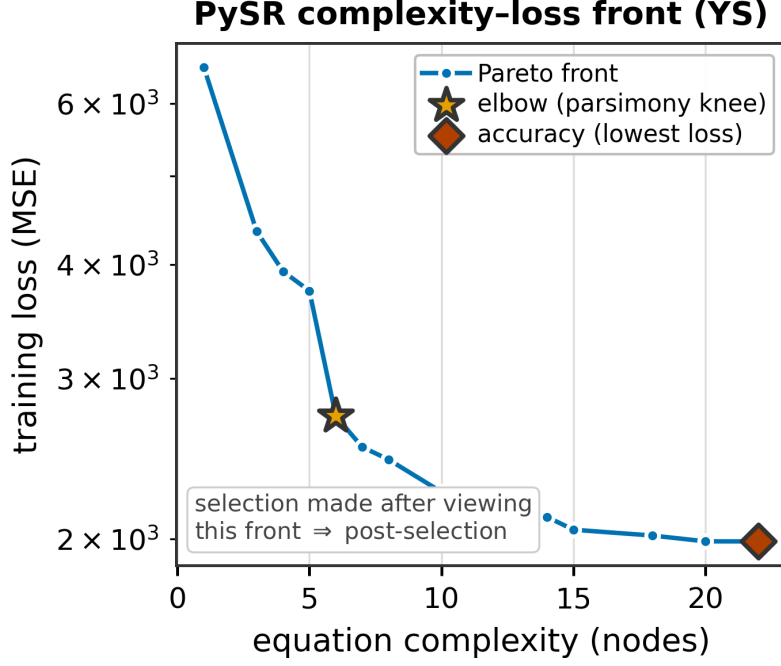


Figure S12: Archived PySR complexity–accuracy fronts. These descriptive plots show the complete-data search. Choosing an equation from them is part of the post-selection boundary.

PySR BIC counts the fitted numerical constants but not the many variables, operators, and structures that were attempted. We therefore do not compare it with the BIC of a pre-specified linear model set. During external evaluation, the protected functions are $\sqrt{|x|}$ and $\log(|x| + 10^{-12})$. The printed equation must include these protections; otherwise the displayed and evaluated forms are different. Protection keeps a square root real but does not remove a pole when that root is in a denominator. The P3–O3 accuracy equation contains $\text{SD}_{\text{grain}}/\sqrt{|\Delta S_{\text{mix}} - 8.2739|}$. The internal minimum $\Delta S_{\text{mix}} = 8.3637$ is only 0.0898 above the singular value, so this expression must not be extrapolated without a stated descriptor-domain check.

SISSO Full has the form

$$\sigma_y = 120.5 \frac{\sigma_r^2}{r_{\text{range}}} + 9356 \frac{d^{-1/2}}{\Delta S_{\text{mix}}} + 1134 \frac{\sigma_\chi^2}{\delta_\mu} - 43.3. \quad (\text{S7})$$

The δ_μ denominator becomes small when the constituents have similar shear moduli. The aggregate external RMSE is 420.6 MPa. The bounded SISSO Robust variant removes this ratio and gives 162.9 MPa. This 61% reduction in RMSE does not establish deployability because $R^2 = -1.37$.

S6.3 Corrected HV fixed form

The refit hardness form is summarized in Table S17. The previously printed equation had two coefficient signs opposite to those in the fitted model. When evaluated as printed, it gives full-data $R^2 = -29.81$ and RMSE 185.4 HV. The corrected full-data fit is

$$H_V = 222.306 + 84.479 \frac{6.93 - d}{\text{SD}_{\text{grain}}} - 1.0046 \frac{\Delta H_{\text{mix}}}{t_{\text{hold}}^2}. \quad (\text{S8})$$

Table S17: Performance of the corrected HV fixed form when its coefficients are refit within every fold.

Protocol	R^2	RMSE (HV)
Full fit	0.744	16.90
5-fold	0.725	17.50
LOO	0.727	17.46
LOBO	0.636	20.16

A 10,000-resample bootstrap gives $c_0 = 221.82 \pm 9.64$, $c_1 = 84.12 \pm 7.34$, and $c_2 = -1.006 \pm 0.106$ as the bootstrap mean $\pm$ SD. Here c_0 and c_1 have units of HV, and c_2 has units of $\text{HV} \cdot \text{h}^2 \cdot \text{mol kJ}^{-1}$. The 95% interval excludes zero for each coefficient. These intervals are conditional on the selected equation structure and therefore do not include structure-selection uncertainty.

The fixed form predicts held-out batches within the combined program, but its structure was selected after the full-data search fronts were examined. Hold time is 0.5 h for every B-campaign record and is 1, 2, 4, or 8 h in Campaign C. It separates campaign with $\text{AUC} = 1.000$, and $\Delta H_{\text{mix}}/t_{\text{hold}}^2$ gives $\text{AUC} = 0.970$. No hold-time effect is identifiable within Campaign B. By comparison, the grain-width term $(6.93 - d)/\text{SD}_{\text{grain}}$ gives $\text{AUC} = 0.533$ and is nearly campaign-neutral. The hold-time term is therefore a campaign proxy in this dataset, whereas the grain-width term is a separate post-selected microstructure hypothesis. The equation is a prospective hypothesis, not yet a deployable hardness law.

The zero at $d = 6.93 \mu\text{m}$ is not a singularity. Singularities occur when $\text{SD}_{\text{grain}} = 0$ or $t_{\text{hold}} = 0$. In the observed data, every d exceeds $14.66 \mu\text{m}$, $\text{SD}_{\text{grain}} \geq 8.32 \mu\text{m}$, and $t \geq 0.5 \text{ h}$. The audit script evaluates four registered equations over the observed and external values. It does not sweep every closed form over a multivariate convex hull.

S7 Literature stress test by evidence class

The literature table combines records with different target provenance. Class E1 contains 54 direct strength values: 48 records from the Borg/Citrine compilation [18] and six compression records for CrFeNi from Schneider and Laplanche [19]. These records represent only 18 exact compositions and mix sources and test modes. The legacy extraction restricted d to 3–500 μm , removed YS values above 800 MPa, and omitted one large-grain CrFeNi record because it departed from the expected Hall–Petch trend. Because these rules use outcome or relationship information, E1 is a heterogeneous stress test rather than an unbiased external benchmark.

Class E2 contains three YS values reconstructed from published Hall–Petch fits [8]. Class E3 contains 25 YS proxies converted from HV [20] using C_{eff} inferred from the present dataset. E2 and E3 are therefore model-derived evidence, not independent measurements of the target. With only three values, the E2 R^2 entries in Table S18 are descriptive and are not interpretable as generalization estimates. These evidence classes do not change the five model families, and they are never pooled to inflate the apparent number of direct external measurements.

Table S18: Literature stress-test metrics separated by evidence class. Negative R^2 on E1 means the model does not beat the E1 mean baseline.

Class	Model family and form	n	R^2	RMSE (MPa)
E1 measured YS	F3 M3	54	-0.508	149.5
E1 measured YS	F5 SISSO Robust	54	-0.726	160.0
E1 measured YS	F5 SISSO Full	54	-5.084	300.3
E2 HP-derived	F3 M3	3	0.677	49.0
E2 HP-derived	F5 SISSO Robust	3	0.849	33.5
E3 HV-converted	F3 M3	25	-2.157	96.9
E3 HV-converted	F5 SISSO Robust	25	-9.642	177.9
E1-E3	F3 M3	82	-0.580	132.9
E1-E3	F5 SISSO Robust	82	-1.373	162.9
E1-E3	F5 SISSO Full	82	-14.813	420.6

The direct subset is the primary literature result, and no model beats its mean baseline. Aggregate RMSE values can compare numerical robustness on the same heterogeneous compilation. They cannot establish deployability or unbiased external accuracy. A future external benchmark should be defined before outcomes are inspected and should use consistent tensile definitions. It should also include complete metadata for composition, processing, phase, grain statistics, temperature, and test mode.

S8 Hardness–yield correspondence

Tabor’s empirical Vickers relation is written as $H \simeq 3\sigma_f(\varepsilon_r)$ with $\varepsilon_r \approx 0.08$ [21]. Instrumented-indentation formulations, including that of Dao et al. [22], use different definitions of representative strain. The value 0.08 is therefore not universal. With $C_{\text{eff}} = H/\sigma_y$ and a Hollomon assumption between 0.2% and 8% strain,

$$n_{\text{app}} = \frac{\ln(C_{\text{eff}}/3)}{\ln(0.08/0.002)} = \frac{\ln(C_{\text{eff}}/3)}{\ln 40}. \quad (\text{S9})$$

The dataset gives $C_{\text{eff}} = 5.13 \pm 1.36$ and $n_{\text{app}} \approx 0.15 \pm 0.07$, where the latter uncertainty is propagated from the reported C_{eff} summary. Offset yield is not a measured flow stress at exactly 0.002 true plastic strain, and the calculation assumes Hollomon behavior. The result is therefore an apparent index rather than a measured tensile exponent.

Global Spearman $\rho(\text{HV}, \text{YS}) = 0.46$, while the per-batch values range from 0.70 to 0.95. The association becomes weaker but does not reverse. The revised paper therefore uses “aggregation-induced attenuation” rather than “Simpson’s paradox.” The partial rank correlation after controlling for d is 0.24. This single adjustment does not identify V, or any other composition variable, as a causal scrambler. We do not recommend the fitted V correction without prospective validation.

The archived descriptive regression is

$$\log C_{\text{eff}} = 1.36 + 0.10 \log d - 2.06x_V, \quad (\text{S10})$$

where d is entered in the numerical units used for fitting. This regression describes an association in the present campaigns (Fig. S13). It is neither dimensionally invariant nor mechanistically identified. Agreement at the top of the distribution also depends on the threshold. Two of the five hardest records are among the five strongest; the overlap increases to 8/10 and 11/20 for broader thresholds. These results support using HV for ranking only within a narrow composition window with similar processing.

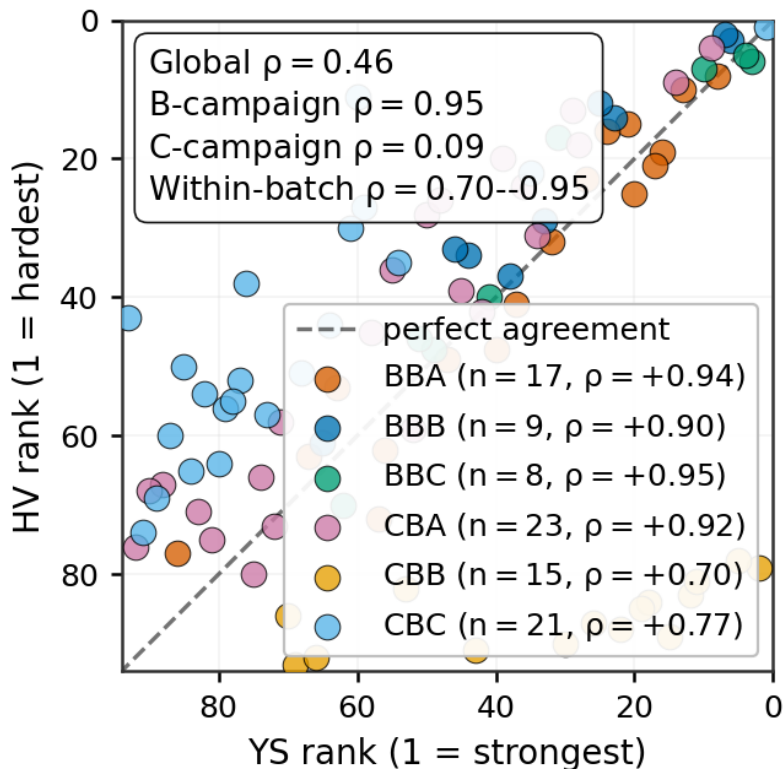


Figure S13: HV rank versus YS rank by batch. Strong ordering within each batch becomes weaker after pooling because composition and processing coverage differ among batches.

S9 Reproducibility across model families

The scripts `verified_analysis.py`, `main2_validation.py`, and `verify_sdgrain_models.py` in `paper_gpt/` generate the revision-specific audit files. The second script writes `verified_results/main2_grouped_validation.csv`, which contains the three internal protocols used in the main paper: 5-fold, LOO, and LOBO. The third writes `verified_results/sdgrain_m15_validation.csv`, which independently refits M3, the additive SD control, and M15. Other verified files contain the fold-contained Family 2 PCA predictions, corrected Family 5 hardness coefficients and bootstrap intervals, the cross-batch composition audit, the E1–E3 external evidence classes, the singularity audit, and a method-claim audit.

Family 3 M15 OLS coefficients, information criteria, and grouped-CV scores are reproducible from the prepared dataset. Its earlier Bayesian stacking weight and PSIS-LOO comparison remain archival because that probabilistic analysis was not regenerated. The Family 4 nested ARMOTE-CV table remains archival because the raw folds and optimization objects are absent. Family 5 PySR remains exploratory until equation discovery is repeated within every outer training split. Full-data loadings, feature rankings, and symbolic search fronts are descriptive and should not be substituted for held-out performance.

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